problems/MHD/orszagTangBormanis2024

DynSGS and positivity setup, as coded on branch `sm/MHD` (commit `dcc3653`)

How to read the underlines. Solid underline: the setting (or the code path) is *identical in every case of `problems/MHD` to which it applies* — all nine cases for general settings, the eight `DSGS_MHD` cases for DynSGS settings, the three `:positivity_method => "conservative"` cases for the element limiter. Dashed underline: shared by a *majority* of those cases but not all of them. No underline: specific to this case or to a minority. The cross-case comparison is in `summary.pdf` of this directory.

Scope. Read off `user_inputs.jl`, the case files and the kernels they reach on `sm/MHD`, with every `JEXPRESSO_OT_*` override unset. Of the eight DynSGS cases this is the only one that runs the *physical* form of the operator (`:dsgs_conserved` left at its default `false`).

1 The problem as set up

Deck key / item	Value as run	Source / remark
Equations	2D ideal GLM-MHD, state $(\rho, \rho u, \rho v, E, \rho w, B_x, B_y, B_z, \psi)$	`user_flux.jl`
EOS	$p = (\gamma-1)(E - \frac{1}{2}\lvert\mathbf{m}\rvert^2/\rho - \frac{1}{2}\lvert\mathbf{B}\rvert^2 - \frac{1}{2}\psi^2)$, $\gamma = 5/3$	`pressure_mhd`, no floor
Domain, mesh	$[0,1]^2$, doubly periodic; `OT_32x32_periodic.msh` refined once by p4est $\Rightarrow 64 \times 64$ quads	`:linitial_refine => true`, `:init_refine_lvl => 1`
Discretization	CG spectral elements, LGL nodes, lumped mass; $N = 4$ (256 unique points per direction)	
Time integration	`CarpenterKennedy2N54`, fixed Δt, $\Delta t = 2 \times 10^{-4}$, $t_{end} = 1$	
Variables	`TOTAL()`, `"energy"`	
Initial data	$\rho = \gamma^2/(4\pi)$, $p = \gamma/(4\pi)$, $\mathbf{v} = (-\sin 2\pi y, \sin 2\pi x)$, $\mathbf{B} = B_0(-\sin 2\pi y, \sin 4\pi x)$, $B_0 = 1/\sqrt{4\pi}$, $w = B_z = \psi = 0$	`initialize.jl`
GLM cleaning	$c_h = \max(\lvert\mathbf{v}\rvert + c_f)$ of the initial data, constant; $S_\psi = -(c_h/0.18)\,\psi$	`initialize.jl`, `user_source.jl`
Sources	`:lsource => true` (ψ damping only)	
Boundary conditions	periodic (free-slip fallback in `user_bc.jl` only for non-periodic tags)	
Other switches	`:lfilter`, `:lkep`, `:entropy_variables`, `:lrichardson`, `:ladapt` false	the filter is off on purpose (deck comment)

2 DynSGS: deck keys and effective values

Deck key / item	Value as run	Source / remark
`:visc_model`	`DSGS_MHD()`, `:lvisc => true`	`SGS.jl:1598`
`:`μ	all 1	

Deck key / item	Value as run	Source / remark
`:dsgs_sensor`	`"legacy"`	assembled RHS against a fixed BDF2
`:dsgs_CR`, `:dsgs_Cmax`	1.0, 0.5	
`:dsgs_Cmin`	0 (default)	
`:dsgs_gamma`	$= \gamma_{mhd} = 5/3$	
`:dsgs_Prt`	1.0	enters the energy coefficient here
`:dsgs_rel`	1.0	`JEXPRESSO_OT_REL`
`:dsgs_norms`	`"domain"`	`JEXPRESSO_OT_NORMS`
`:dsgs_conserved`	**false** (default)	physical form: the only such case
`:dsgs_nazarov_energy`	true	$\kappa = \bar{\rho}\nu/\,\mathrm{Pr}$ on ∇T
`:dsgs_nodal_rho`	false (default)	element-mean $\bar{\rho}_K$ in the dynamic coefficients
`:dsgs_fast_floors`, `:dsgs_thermal_residual`, `:dsgs_ref_weight`	false (defaults)	
`:dsgs_hold_steps`	2	`JEXPRESSO_OT_HOLD`
`:dsgs_cutoff`, `:dsgs_freeze_stage`, `:dsgs_reference`	0, false, false (defaults)	
`:ldsgs_nodal`	false (element form)	
`:dsgs_local_rel`	1.0 (default)	unused (domain norms)

3 DynSGS as it runs for this case

$\Delta_K^{elem} = 1/64$, $\Delta_K = \Delta_K^{elem}/(N+1) = 1/320$; $h = \Delta t$.

1. **History and hold.** Rotation once per step (`rhs.jl:936`); two held steps, during which $\nu = 0$ ($C_{\min} = 0$).
2. **Legacy residual** (`rhs.jl:960-964`, `_dsgs_residual_rhs!`): at *every* stage, with the stage value q_I,

$$R_I^s = \Big| \frac{3q_I^s - 4q_I^{n,s} + q_I^{n-1,s}}{2h} - \big(M^{-1}\mathcal{R}_{inv}\big)_I^s \Big|,$$

   $\mathcal{R}_{inv}$ the inviscid right-hand side (flux + ψ source) assembled over ranks and periodic copies (`DSS_global_RHS!` on a copy), M the lumped global mass. The value is written into the element layout so the kernel sees the same number from every element sharing I. No Dirichlet zeroing on this path (and none needed: periodic).
3. **Normalization** (domain): $n^s = \max(S^s, \sigma^s) + 10^{-16}$ with the mean-state floors $\sigma^\rho = \bar{\rho}$, $\sigma^{\mathbf{m}} = \bar{\rho}\bar{c}$, $\sigma^E = \bar{\rho}\bar{c}^2$, $\sigma^{\mathbf{B}} = \sqrt{\bar{\rho}}\,\bar{c}$, sound speed $\bar{c} = \sqrt{\gamma\bar{p}/\bar{\rho}}$. ρ is uniform at $t = 0$, so its residual is first measured against $\bar{\rho}$.
4. **Ratio, cap, coefficient.** $\mathcal{R}_K = \max_{I\in K} \max_{s\le 8} R_I^s/n^s$; $w_K = \max_I(|\mathbf{v}_I| + \sqrt{\max(\gamma p_I/\rho_I + |\mathbf{B}_I|^2/\rho_I, 0)})$; $\nu_K = \max(0, \min(0.5\,\Delta_K w_K,\ \Delta_K^2 \mathcal{R}_K))$, no floor, one value per element, recomputed at every stage.
5. **Slot coefficients** (physical branch, `SGS.jl:1889-1900`), with $\bar{\rho}_K$ the plain average of nodal ρ over the $(N+1)^2$ nodes of K:

$$\mu^\rho = \nu_K, \quad \mu^{\rho u} = \mu^{\rho v} = \mu^{\rho w} = \bar{\rho}_K\nu_K, \quad \mu^E = \bar{\rho}_K\nu_K/\,\mathrm{Pr} = \bar{\rho}_K\nu_K, \quad \mu^B = \nu_K, \quad \mu^\psi = \nu_K.$$

4 The viscous operator

`user_primitives.jl` returns $(\rho, u, v, T{=}p/\rho, w, B_x, B_y, B_z, \psi)$, and with `lcons = false` `_expansion_visc!` (`rhs.jl:2943`) assembles:

Equation	Viscous flux $\mathbf{F}_V$	Coefficient
ρ	$\nu\nabla\rho$	mass diffusion on
$\rho u,\ \rho v$	$\boldsymbol{\tau} = \bar{\rho}\nu\big(\nabla\mathbf{v} + \nabla\mathbf{v}^T - \frac{2}{3}(\nabla\cdot\mathbf{v})I\big)$, $\mathbf{v} = (u, v)$	$\bar{\rho}\nu$
ρw	$\bar{\rho}\nu\nabla w$	$\bar{\rho}\nu$
E	$\bar{\rho}\nu\nabla T + \boldsymbol{\tau}\cdot\mathbf{v} + \nu\sum_k B_k\nabla B_k$	heat + viscous work + resistive work
B_k	$\nu\nabla B_k$	$\eta = \nu$
ψ	$\nu\nabla\psi$	

The work terms (rhs.jl:3076-3122) return the in-plane kinetic and the magnetic energy removed by the stress and the induction Laplacian to E. The ρw and ψ dissipations have no matching work term ($w \equiv 0$ in this problem).

5 Positivity

- :lpositivity false, no stage limiter, no state modification.
- No flux-only floor: pressure_mhd of this case is unguarded, so a negative p reaches the flux as is.
- Admissibility therefore rests entirely on the DynSGS dissipation; the deck records why the Boyd–Vandeven filter is kept off (it filters ρ, $\rho\mathbf{v}$, E independently and can drive $p < 0$).

6 Observations: code versus comments

1. The comment above :dsgs_norms says "the code-wide default is "rank""; the default in mod_inputs.jl:1161 is "domain", which is also what the deck sets.
2. The comments say $\Delta t = 5 \times 10^{-4}$ and ":nop => 4 on the 32×32 grid gives the 128^2 of the paper"; the deck sets $\Delta t = 2 \times 10^{-4}$ and refines the mesh once (64×64 elements, 256^2 points).
3. :output_dir points to a cluster scratch path (/scratch/smarras/...).

Appendix: code map (branch sm/MHD, commit dcc3653)

Mechanism	Where (file:line)
Deck defaults of every :dsgs_* key	src/io/mod_inputs.jl:1085-1376
Deck defaults of :lpositivity, floors, report	src/io/mod_inputs.jl:456-477
:positivity_method; rebuild of the RK solver	src/io/mod_inputs.jl:828-838
DynSGS history rotation, stage weights, hold, freeze	src/kernel/operators/rhs.jl:927-984
Stencil selection _dsgs_stencil	rhs.jl:141
Hold length _dsgs_hold_steps ($\max(2, H) + 1$ rotations)	rhs.jl:182
Residual right-hand side (element / legacy / Dirichlet)	rhs.jl:188-280; legacy fill rhs.jl:282; boundary zeroing rhs.jl:313, 346
1D DSGS_MHD dispatch	rhs.jl:1398-1470
2D DSGS_MHD dispatch (sets dsgs_ref_weight, dsgs_split_energy, dsgs_nodal_rho)	rhs.jl:1517-1610
2D viscous assembly _viscous_rhs_el_2d_dsgs!	rhs.jl:1796
2D weak viscous flux _expansion_visc! (conserved flag, ref. weight, split energy, $\tau \cdot \mathbf{v}$, resistive work)	rhs.jl:2943-3210 (lcons 2959, weight 2978, split 3050, $\tau \cdot \mathbf{v}$ 3076, $\eta\,\mathbf{B} \cdot \nabla\mathbf{B}$ 3091)
1D weak viscous flux	rhs.jl:2823; element loop rhs.jl:1321
2D DSGS_MHD element kernel	src/kernel/physics/SGS.jl:1598-1912
1D DSGS_MHD element kernel	SGS.jl:1929-2120
Nodal kernels (off in every MHD deck; BW can switch it on with JEXPRESSO_BW_NODAL=1)	SGS.jl:2144 (1D), SGS.jl:2515 (2D)
Normalization floor _dsgs_denom, :dsgs_rel multiplier	SGS.jl:676-679; set per call in rhs.jl:192
Smoothness cutoff _dsgs_cut	SGS.jl:703
SGS_diffusion(::DSGS_MHD) (nodal-ρ switch)	SGS.jl:547-569
Element length Δ_K^{elem} (2D: min of corner extents)	src/kernel/mesh/mesh.jl:6430-6457; 1D Δx: mesh.jl:5863
uprimitive width ($n_{eqs} + 2$ in 2D, $n_{eqs} + 1$ in 1D)	src/kernel/globalStructs.jl:139-153
Realizability repair ("repair", unused by the MHD decks)	src/kernel/positivity/Positivity.jl:294; driver positivity_driver.jl:124; called in rhs.jl:821
Conservative element limiter ("conservative")	Positivity.jl:433-559; driver positivity_driver.jl:245-337
Main solve call (saveat non-empty ⇒ calck = true)	src/kernel/solvers/TimeIntegrators.jl:855

Stage-limiter calls in `CarpenterKennedy2N54`: with `williamson_condition = false` the limiter is called before the right-hand side of stages 2–5 and once after stage 5 (OrdinaryDiffEq v6.90/v6.95, `low_storage_rk_perform_step.jl`). In OrdinaryDiffEq `calck = true` forces `williamson_condition = false` in the 2N cache, and Jexpresso's `solve` passes a non-empty `saveat`, so this holds for every deck that hands the solver a stage limiter. (Jexpresso ships no `Manifest.toml`; the package version is whatever the environment resolves, and the check above was made against v6.90/v6.95.)